

Assignment 6

Array-Based Programs in Java

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Objective

The objective of this set of programs is to understand and implement basic array operations in Java, including traversal, searching, reversing, and counting occurrences of elements, using loops and conditional statements.

1. Find the Sum of Array Elements

This program reads the size of an array and its elements from the user, then calculates the sum of all elements using a for loop.

Java Program

```
import java.util.Scanner;

public class SumOfArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        for (int i = 0; i < n; i++) {
            sum = sum + arr[i];
        }

        System.out.println("Sum of array elements = " + sum);

        sc.close();
    }
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements: 10 20 30 40 50
Sum of array elements = 150
```

2. Find the Largest Element in an Array

This program reads an array from the user and finds the largest element by comparing each element with a running maximum value.

Java Program

```
import java.util.Scanner;

public class LargestInArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int largest = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }

        System.out.println("Largest element in the array = " + largest);

        sc.close();
    }
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements: 12 45 7 89 34
Largest element in the array = 89
```

3. Search an Element in an Array

This program reads an array and a key element from the user, then performs a linear search to find the index at which the key is present, if it exists.

Java Program

```
import java.util.Scanner;

public class SearchElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter the element to search: ");
        int key = sc.nextInt();

        int position = -1;
        for (int i = 0; i < n; i++) {
            if (arr[i] == key) {
                position = i;
                break;
            }
        }

        if (position != -1) {
            System.out.println(key + " found at index " + position);
        } else {
            System.out.println(key + " not found in the array");
        }

        sc.close();
    }
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements: 10 20 30 40 50
Enter the element to search: 30
30 found at index 2
```

4. Reverse an Array

This program reads an array from the user and reverses it in place by swapping elements from both ends towards the center using two pointers.

Java Program

```
import java.util.Scanner;

public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int start = 0, end = n - 1;
        while (start < end) {
            int temp = arr[start];
            arr[start] = arr[end];
            arr[end] = temp;
            start++;
            end--;
        }

        System.out.print("Reversed array: ");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();

        sc.close();
    }
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements: 1 2 3 4 5
Reversed array: 5 4 3 2 1
```

5. Count Occurrence of an Element in an Array

This program reads an array and a key element from the user, then counts how many times the key appears in the array using a loop and a counter variable.

Java Program

```
import java.util.Scanner;

public class CountOccurrence {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter the element to count: ");
        int key = sc.nextInt();

        int count = 0;
        for (int i = 0; i < n; i++) {
            if (arr[i] == key) {
                count++;
            }
        }

        System.out.println("Occurrence of " + key + " in the array = " + count);

        sc.close();
    }
}
```

Output

```
Enter the number of elements: 7
Enter 7 elements: 2 3 2 4 2 5 2
Enter the element to count: 2
Occurrence of 2 in the array = 4
```